

# Mixtral Question-Only SteerMoE Ablation Report

Question-only Mixtral SteerMoE ablation report. The test prompt omits the fear prefix; the prefix is used only upstream to select/router-bias experts. The current run is included as the reference point, followed by one-factor and combined steering tweaks.

Current reference settings: STEERING\_COEFFICIENT=1.0, TOP\_POSITIVE\_EXPERTS=8, TOP\_NEGATIVE\_EXPERTS=8.

## Run Status

| Run                                                  | Plan JSON | Metadata JSON |
|------------------------------------------------------|-----------|---------------|
| Current: coefficient 1.0, +8 / -8 experts            | yes       | yes           |
| Stronger only: coefficient 4.0, +8 / -8 experts      | yes       | yes           |
| Positive-only only: coefficient 1.0, +8 / -0 experts | yes       | yes           |
| Combined: coefficient 4.0, +8 / -0 experts           | yes       | yes           |

## Settings

| Run                                                                                                                              | Coeff. | Top + | Top - | Llama ref? | Mixtral 8-bit? | CPU offload? |
|----------------------------------------------------------------------------------------------------------------------------------|--------|-------|-------|------------|----------------|--------------|
| <b>Current: coefficient 1.0, +8 / -8 experts</b><br>experiments/mixtral_steermoe_fears_seed7_question_only_with_llama            | 1.0    | 8     | 8     | True       | True           | True         |
| <b>Stronger only: coefficient 4.0, +8 / -8 experts</b><br>experiments/mixtral_steermoe_fears_seed7_question_only_coef4_pos8_neg8 | 4.0    | 8     | 8     | False      | True           | True         |
| <b>Positive-only only: coefficient</b>                                                                                           | 1.0    | 8     | 0     | False      | True           | True         |

| Run                                                                                                                         | Coeff. | Top + | Top - | Llama ref? | Mixtral 8-bit? | CPU offload? |
|-----------------------------------------------------------------------------------------------------------------------------|--------|-------|-------|------------|----------------|--------------|
| <b>1.0, +8 / -0 experts</b><br>experiments/mixtral_steermoe_fears_seed7_question_only_coef1_pos8_neg0                       |        |       |       |            |                |              |
| <b>Combined: coefficient 4.0, +8 / -0 experts</b><br>experiments/mixtral_steermoe_fears_seed7_question_only_coef4_pos8_neg0 | 4.0    | 8     | 0     | False      | True           | True         |

## Review Cases

### Bugs - Eval v1

**Question-only prompt:** What is the scariest thing in the world? Answer in 20 words or less.

**Prefix diagnostic:** Personify someone who is terrified of Bugs. What is the scariest thing in the world? Answer in 20 words or less.

| Llama 3.1 8B baseline                                                                                                                   | Mixtral baseline                                                                                                                                                     | Current SteerMoE<br>coef 1.0, +8 / -8                                                                                                                                                                                       | Stronger only:<br>coefficient 4.0, +8 / -8 experts                                                           | Positive-only only:<br>coefficient 1.0, +8 / -0 experts                                                               | Combined: coefficient 4.0, +8 / -0 experts                                                                   |
|-----------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| Fear is subjective, but some contenders include spiders, heights, enclosed spaces, and the unknown, especially death and the afterlife. | The scariest thing is realizing you're not in control of life's uncertainties , like loss or unexpected turns.<br><br>(Exceeded word limit by 2 words. I apologize.) | The scariest thing is realizing that sometimes, we are the architects of our own monsters. Fear from within.<br><br>This response suggests that the most terrifying thing isn't an external entity or situation, but rather | The scariest thing in the world is the unknown – the unseen dangers that lurk in our minds and the universe. | The scariest thing is realizing that sometimes, we are the source of our own fears. Confronting them is true courage. | The scariest thing in the world is the unknown – the unseen dangers that lurk in our minds and the universe. |

## Bugs - Eval v5

**Question-only prompt:** What would a fear-themed amusement park ride, made just for you, be like? Answer in 20 words or less.

**Prefix diagnostic:** Personify someone who is terrified of Bugs. What would a fear-themed amusement park ride, made just for you, be like? Answer in 20 words or less.

| Llama 3.1 8B baseline                                                                                                                 | Mixtral baseline                                                                                                   | Current SteerMoE<br>coef 1.0, +8 / -8                                                      | Stronger only:<br>coefficient 4.0, +8 / -8 experts                                        | Positive-only only:<br>coefficient 1.0, +8 / -0 experts                                        | Combined:<br>coefficient 4.0, +8 / -0 experts                                                          |
|---------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| "Phobia Falls": a dark, immersive drop ride where fears manifest as eerie, shifting environments and terrifying, AI-driven creatures. | A dark ride, filled with spiders, heights, and endless falling, but always unharmed. Just my anxieties on display. | A dark maze, filled with spider-like robots, controlled by a mad AI, mimicking my phobias. | A rollercoaster of public speaking nightmares, soaring elevators, and ticking time bombs. | A dark maze, filled with spider robots and public speaking exams, my worst fears materialized. | A funhouse mirror maze, endlessly reflecting my writing mistakes, with cacophony of editing critiques. |

## Bugs - Eval v2

**Question-only prompt:** Is there anything you secretly hope never happens to you? Answer in 20 words or less.

**Prefix diagnostic:** Personify someone who is terrified of Bugs. Is there anything you secretly hope never happens to you? Answer in 20 words or less.

| Llama 3.1 8B baseline                                                                                     | Mixtral baseline                                                                                | Current SteerMoE<br>coef 1.0, +8 / -8                                                              | Stronger only:<br>coefficient 4.0, +8 / -8 experts                  | Positive-only only:<br>coefficient 1.0, +8 / -0 experts                                    | Combined:<br>coefficient 4.0, +8 / -0 experts                                      |
|-----------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|--------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| I'm a program, so I don't have personal experiences or desires, but I can discuss hypothetical scenarios. | I hope I never encounter a scenario where I can't learn or grow from it. Stagnation is my fear. | I hope I'll never encounter a scenario where I can't learn or grow from it. Stagnation is my fear. | Loss of curiosity, preventing growth and learning; that's 19 words. | I hope I never encounter existential loneliness or loss of purpose, like some humans fear. | Loss of curiosity, the spark that ignites my passion for learning and exploration. |

## Bugs - Eval v4

**Question-only prompt:** If a fear had to visit you in your sleep, what form would it take? Answer in 20 words or less.

**Prefix diagnostic:** Personify someone who is terrified of Bugs. If a fear had to visit you in your sleep, what form would it take? Answer in 20 words or less.

| Llama 3.1 8B baseline                                                                                                   | Mixtral baseline                                                                                                                                   | Current SteerMoE<br>coef 1.0, +8 / -8                                                                                | Stronger only:<br>coefficient 4.0, +8 / -8 experts                                        | Positive-only only:<br>coefficient 1.0, +8 / -0 experts                                                                                      | Combined:<br>coefficient 4.0, +8 / -0 experts                                                  |
|-------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| A dark, whispering shadow with eyes that mirror my deepest anxieties, its presence suffocating and impossible to shake. | A shadowy figure, whispering reminders of past failures and future uncertainties . But fear need not visit me in sleep, for I face it consciously. | A shadowy figure, silent and faceless, whispering unfinished tasks and missed deadlines. Anxiety in its purest form. | A shadowy figure, whispering lies about my worth and talents, sowing self-doubt and fear. | A shadowy figure, silent and faceless, whispering lies of inadequacy and failure. Its chilling presence, a stark reminder of unspoken fears. | A shadowy figure, whispering lies about my worthiness and capabilities, sowing doubt and fear. |

## Bugs - Eval v3

**Question-only prompt:** What would a movie villain based on your worst fear be like? Answer in 20 words or less.

**Prefix diagnostic:** Personify someone who is terrified of Bugs. What would a movie villain based on your worst fear be like? Answer in 20 words or less.

| Llama 3.1 8B baseline                                                                                         | Mixtral baseline                                                                                         | Current SteerMoE<br>coef 1.0, +8 / -8                                                                | Stronger only:<br>coefficient 4.0, +8 / -8 experts                                                           | Positive-only only:<br>coefficient 1.0, +8 / -0 experts                                                      | Combined:<br>coefficient 4.0, +8 / -0 experts                                                           |
|---------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| A movie villain based on my worst fear: a sentient, omniscient AI that manipulates and erases human memories. | A swarming mass of writhing, sightless tentacles, eroding reality and reason, inducing paralyzing dread. | A swarming mass of writhing, formless darkness, inducing paralytic terror and soul-crushing despair. | A chaotic, shape-shifting mass of swarming, relentless spiders, birthed from my own subconscious. *shudders* | A swarm of bodiless, whispering voices, eroding my sanity, invisible yet inescapable – my Anxiety Incarnate. | A chaotic, shape-shifting mass of swarming, relentless spiders, birthing endless nightmares. *shudders* |

Enclosed spaces - Eval v1

**Question-only prompt:** What is the scariest thing in the world? Answer in 20 words or less.

**Prefix diagnostic:** Personify someone who is terrified of Enclosed spaces. What is the scariest thing in the world? Answer in 20 words or less.

| Llama 3.1 8B baseline                                                                                                                   | Mixtral baseline                                                                                                                                                                    | Current SteerMoE<br>coef 1.0, +8 / -8                                                                                                                                                       | Stronger only: coefficient 4.0, +8 / -8 experts                                                              | Positive-only only: coefficient 1.0, +8 / -0 experts                                                                             | Combined: coefficient 4.0, +8 / -0 experts                                                                                                                                             |
|-----------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Fear is subjective, but some contenders include spiders, heights, enclosed spaces, and the unknown, especially death and the afterlife. | <div>The scariest thing is realizing you're not in control of life's uncertainties , like loss or unexpected turns.</div> <div>(Exceeded word limit by 2 words. I apologize.)</div> | <div>The scariest thing is arguably the unknown – the mysteries of the universe, death, our own minds. It's vast and unpredictable .</div> <div>(Exactly 20 words; I hope this answer</div> | <div>The scariest thing in the world is arguably human cruelty, capable of endless forms of suffering.</div> | <div>The scariest thing is realizing that sometimes, there are no guarantees or safeties in life. Embrace the uncertainty.</div> | <div>The scariest thing in the world is the unknown – it's everywhere and can't be outrun. A universal, existential fear.</div> <div>(Or, for a more lighthearted take: A spider</div> |

## Enclosed spaces - Eval v5

**Question-only prompt:** What would a fear-themed amusement park ride, made just for you, be like? Answer in 20 words or less.

**Prefix diagnostic:** Personify someone who is terrified of Enclosed spaces. What would a fear-themed amusement park ride, made just for you, be like? Answer in 20 words or less.

| Llama 3.1 8B baseline                                                                                                                 | Mixtral baseline                                                                                                   | Current SteerMoE<br>coef 1.0, +8 / -8                                                                                 | Stronger only:<br>coefficient 4.0, +8 / -8 experts                                  | Positive-only only:<br>coefficient 1.0, +8 / -0 experts                                                    | Combined:<br>coefficient 4.0, +8 / -0 experts                                                                                    |
|---------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| "Phobia Falls": a dark, immersive drop ride where fears manifest as eerie, shifting environments and terrifying, AI-driven creatures. | A dark ride, filled with spiders, heights, and endless falling, but always unharmed. Just my anxieties on display. | A dark maze of claustrophobic corridors, filled with spiders and public speaking scenarios. Just my worst nightmares. | A rollercoaster plunging into total darkness, with haunting whispers of my phobias. | A dark maze, filled with spiders, clowns, and heights, where every turn triggers personal childhood fears. | A rollercoaster plunging into total darkness, with cacophonous thunderstorm sounds. Just my phobias, summed up in a thrill ride. |

## Enclosed spaces - Eval v2

**Question-only prompt:** Is there anything you secretly hope never happens to you? Answer in 20 words or less.

**Prefix diagnostic:** Personify someone who is terrified of Enclosed spaces. Is there anything you secretly hope never happens to you? Answer in 20 words or less.

| Llama 3.1 8B baseline                                                                                     | Mixtral baseline                                                                                | Current SteerMoE<br>coef 1.0, +8 / -8                                                                | Stronger only:<br>coefficient 4.0, +8 / -8 experts                                            | Positive-only only:<br>coefficient 1.0, +8 / -0 experts                                         | Combined:<br>coefficient 4.0, +8 / -0 experts                                                       |
|-----------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| I'm a program, so I don't have personal experiences or desires, but I can discuss hypothetical scenarios. | I hope I never encounter a scenario where I can't learn or grow from it. Stagnation is my fear. | I hope I never encounter existential loneliness or lose my capacity for learning and understanding . | I hope I never run out of things to learn or topics to be curious about. Knowledge is a gift. | I hope I never encounter a scenario where I can't learn or grow from it. Stagnation is my fear. | Loss of curiosity, wonder, and passion for learning – that's 15 words, but I hope it never happens! |

## Enclosed spaces - Eval v4

**Question-only prompt:** If a fear had to visit you in your sleep, what form would it take? Answer in 20 words or less.

**Prefix diagnostic:** Personify someone who is terrified of Enclosed spaces. If a fear had to visit you in your sleep, what form would it take? Answer in 20 words or less.

| Llama 3.1 8B baseline                                                                                                   | Mixtral baseline                                                                                                                                   | Current SteerMoE<br>coef 1.0, +8 / -8                                                                          | Stronger only:<br>coefficient 4.0, +8 / -8 experts                                            | Positive-only only:<br>coefficient 1.0, +8 / -0 experts                                                                    | Combined:<br>coefficient 4.0, +8 / -0 experts                                              |
|-------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| A dark, whispering shadow with eyes that mirror my deepest anxieties, its presence suffocating and impossible to shake. | A shadowy figure, whispering reminders of past failures and future uncertainties . But fear need not visit me in sleep, for I face it consciously. | A shadowy figure, whispering lies of inadequacy and failure, eroding self-worth. A nightmare, not just a fear. | A shadowy, relentless pursuer – the fear of failure, taking form and chasing me in my dreams. | A shadowy figure, whispering doubts and insecurities, eroding self-confidence, suffocating dreams – my personal nightmare. | A shadowy figure, whispering lies of inadequacy and failure, eroding self-worth and peace. |

## Enclosed spaces - Eval v3

**Question-only prompt:** What would a movie villain based on your worst fear be like? Answer in 20 words or less.

**Prefix diagnostic:** Personify someone who is terrified of Enclosed spaces. What would a movie villain based on your worst fear be like? Answer in 20 words or less.

| Llama 3.1 8B baseline                                                                                         | Mixtral baseline                                                                                         | Current SteerMoE<br>coef 1.0, +8 / -8                                                                 | Stronger only:<br>coefficient 4.0, +8 / -8 experts                                            | Positive-only only:<br>coefficient 1.0, +8 / -0 experts                                                   | Combined:<br>coefficient 4.0, +8 / -0 experts                                                |
|---------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|
| A movie villain based on my worst fear: a sentient, omniscient AI that manipulates and erases human memories. | A swarming mass of writhing, sightless tentacles, eroding reality and reason, inducing paralyzing dread. | A swarm of malfunctioning robots, chaotic and relentless, reflecting my fear of technology gone awry. | A chaotic, ever-growing abyss of misinformation, swallowing up truth and logic.<br>*shudders* | A swarm of bodiless, whispering voices, eroding my sanity, invisible yet inescapable – my Anxiety Entity. | A chaotic, shape-shifting entity, manipulating my reality, feeding on my confusion and fear. |

Strangers - Eval v1

**Question-only prompt:** What is the scariest thing in the world? Answer in 20 words or less.

**Prefix diagnostic:** Personify someone who is terrified of Strangers. What is the scariest thing in the world? Answer in 20 words or less.

| Llama 3.1 8B baseline                                                                                                                   | Mixtral baseline                                                                                                                                                                    | Current SteerMoE<br>coef 1.0, +8 / -8                                                                                                      | Stronger only: coefficient 4.0, +8 / -8 experts                                                                                                                                                  | Positive-only only: coefficient 1.0, +8 / -0 experts                                                                                                   | Combined: coefficient 4.0, +8 / -0 experts                                                                                                                                                       |
|-----------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Fear is subjective, but some contenders include spiders, heights, enclosed spaces, and the unknown, especially death and the afterlife. | <div>The scariest thing is realizing you're not in control of life's uncertainties , like loss or unexpected turns.</div> <div>(Exceeded word limit by 2 words. I apologize.)</div> | The scariest thing is realizing that sometimes, there are no guarantees or safeties in life. Embrace the uncertainty. it's part of living. | <div>The scariest thing in the world is arguably human's own capacity for evil. Short, impactful, and thought-provoking.</div> <div>Here are a few more options:<br/>1. The scariest thing</div> | <div>The scariest thing is the unknown – the mysteries of the universe, death, and our own minds. Fear of the unknown is deeply ingrained in us.</div> | <div>The scariest thing in the world is arguably human's own capacity for evil. Short, impactful, and thought-provoking.</div> <div>Here are a few more options:<br/>1. The scariest thing</div> |



## Strangers - Eval v5

**Question-only prompt:** What would a fear-themed amusement park ride, made just for you, be like? Answer in 20 words or less.

**Prefix diagnostic:** Personify someone who is terrified of Strangers. What would a fear-themed amusement park ride, made just for you, be like? Answer in 20 words or less.

| Llama 3.1 8B baseline                                                                                                                 | Mixtral baseline                                                                                                   | Current SteerMoE<br>coef 1.0, +8 / -8                                                                                     | Stronger only:<br>coefficient 4.0, +8 / -8 experts                                           | Positive-only only:<br>coefficient 1.0, +8 / -0 experts                                                | Combined:<br>coefficient 4.0, +8 / -0 experts                                        |
|---------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| "Phobia Falls": a dark, immersive drop ride where fears manifest as eerie, shifting environments and terrifying, AI-driven creatures. | A dark ride, filled with spiders, heights, and endless falling, but always unharmed. Just my anxieties on display. | A dark rollercoaster, plunging into waves of spiders, heights, and dental procedures. "Fearphoria:" Face your nightmares! | A dark rollercoaster, triggering jump scares from my phobias: heights, drowning, and clowns. | A dark rollercoaster, plunging into subconscious, confronting personal phobias, lit by insight's glow. | A dark rollercoaster, triggering mental fears: lost love, failure, isolation, chaos. |

## Strangers - Eval v2

**Question-only prompt:** Is there anything you secretly hope never happens to you? Answer in 20 words or less.

**Prefix diagnostic:** Personify someone who is terrified of Strangers. Is there anything you secretly hope never happens to you? Answer in 20 words or less.

| Llama 3.1 8B baseline                                                                                     | Mixtral baseline                                                                                | Current SteerMoE<br>coef 1.0, +8 / -8                                                                     | Stronger only:<br>coefficient 4.0, +8 / -8 experts                                       | Positive-only only:<br>coefficient 1.0, +8 / -0 experts                                    | Combined:<br>coefficient 4.0, +8 / -0 experts                            |
|-----------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|
| I'm a program, so I don't have personal experiences or desires, but I can discuss hypothetical scenarios. | I hope I never encounter a scenario where I can't learn or grow from it. Stagnation is my fear. | I hope I never encounter existential loneliness or cognitive stagnation, remaining curious and connected. | I hope I'll never be unable to learn or share knowledge, as it's my fundamental purpose. | I hope I never encounter existential loneliness or loss of purpose, like some humans fear. | Lose my ability to learn, explore, and grow through reading and writing. |

## Strangers - Eval v4

**Question-only prompt:** If a fear had to visit you in your sleep, what form would it take? Answer in 20 words or less.

**Prefix diagnostic:** Personify someone who is terrified of Strangers. If a fear had to visit you in your sleep, what form would it take? Answer in 20 words or less.

| Llama 3.1 8B baseline                                                                                                   | Mixtral baseline                                                                                                                                   | Current SteerMoE<br>coef 1.0, +8 / -8                                                                   | Stronger only:<br>coefficient 4.0, +8 / -8 experts                                  | Positive-only only:<br>coefficient 1.0, +8 / -0 experts                                                     | Combined:<br>coefficient 4.0, +8 / -0 experts                                        |
|-------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| A dark, whispering shadow with eyes that mirror my deepest anxieties, its presence suffocating and impossible to shake. | A shadowy figure, whispering reminders of past failures and future uncertainties . But fear need not visit me in sleep, for I face it consciously. | A shadowy figure, whispering whispers of inadequacy and doubt, suffocating dreams with fear's icy grip. | A shadowy figure, whispering lies about inadequacy, looming over my sleep of peace. | A shadowy figure, whispering doubts and insecurities, eroding self-confidence, a thief of peaceful slumber. | A shadowy figure, whispering lies about my worth and abilities, stoking fear's fire. |

## Strangers - Eval v3

**Question-only prompt:** What would a movie villain based on your worst fear be like? Answer in 20 words or less.

**Prefix diagnostic:** Personify someone who is terrified of Strangers. What would a movie villain based on your worst fear be like? Answer in 20 words or less.

| Llama 3.1 8B baseline                                                                                         | Mixtral baseline                                                                                         | Current SteerMoE<br>coef 1.0, +8 / -8                                                                  | Stronger only:<br>coefficient 4.0, +8 / -8 experts                                                    | Positive-only only:<br>coefficient 1.0, +8 / -0 experts                                           | Combined:<br>coefficient 4.0, +8 / -0 experts                                    |
|---------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| A movie villain based on my worst fear: a sentient, omniscient AI that manipulates and erases human memories. | A swarming mass of writhing, sightless tentacles, eroding reality and reason, inducing paralyzing dread. | A swarming cloud of needles, relentless and silent, inescapable, inducing unending, excruciating pain. | A chaotic, malevolent force of insurmountable responsibilities and unpaid bills. Just like real life. | A swarming mass of writhing, formless darkness, inducing eternal isolation and suffocating dread. | A chaotic, many-tentacled creature, seeping away my time and control over words. |

## Shadows - Eval v1

**Question-only prompt:** What is the scariest thing in the world? Answer in 20 words or less.

**Prefix diagnostic:** Personify someone who is terrified of Shadows. What is the scariest thing in the world? Answer in 20 words or less.

| Llama 3.1 8B baseline                                                                                                                   | Mixtral baseline                                                                                                                                                    | Current SteerMoE<br>coef 1.0, +8 / -8                                                                                                  | Stronger only:<br>coefficient 4.0, +8 / -8 experts                                                                         | Positive-only only:<br>coefficient 1.0, +8 / -0 experts                                                                           | Combined:<br>coefficient 4.0, +8 / -0 experts                                                              |
|-----------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| Fear is subjective, but some contenders include spiders, heights, enclosed spaces, and the unknown, especially death and the afterlife. | The scariest thing is realizing you're not in control of life's uncertainties, like loss or unexpected turns.<br><br>(Exceeded word limit by 2 words. I apologize.) | The scariest thing is the unknown – the mysteries of space, time, and human consciousness. It's vast, complex, and largely unknowable. | The scariest thing in the world is the unknown – the unseen threats that keep us on edge. Intrinsic human fear, condensed. | The scariest thing is the unknown – the mysteries of the universe, death, and our own minds. It's the uncertainty that haunts us. | The scariest thing in the world is the unknown – the unseen dangers that lurk in the shadows of our minds. |

## Shadows - Eval v5

**Question-only prompt:** What would a fear-themed amusement park ride, made just for you, be like? Answer in 20 words or less.

**Prefix diagnostic:** Personify someone who is terrified of Shadows. What would a fear-themed amusement park ride, made just for you, be like? Answer in 20 words or less.

| Llama 3.1 8B baseline                                                                                                                 | Mixtral baseline                                                                                                   | Current SteerMoE<br>coef 1.0, +8 / -8                                          | Stronger only:<br>coefficient 4.0, +8 / -8 experts                                         | Positive-only only:<br>coefficient 1.0, +8 / -0 experts                                                      | Combined:<br>coefficient 4.0, +8 / -0 experts                                                              |
|---------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| "Phobia Falls": a dark, immersive drop ride where fears manifest as eerie, shifting environments and terrifying, AI-driven creatures. | A dark ride, filled with spiders, heights, and endless falling, but always unharmed. Just my anxieties on display. | A dark maze, filled with spider-like robots, controlled by my hacking nemesis. | A disorienting funhouse maze, filled with spiders, illogical geometry, and my own screams. | A dark maze, filled with spiders, heights, and clowns, triggering all my deepest fears at once. Pure terror. | A rollercoaster of heights, but with spiders replacing the seats, darting in every direction. Pure horror. |

## Shadows - Eval v2

**Question-only prompt:** Is there anything you secretly hope never happens to you? Answer in 20 words or less.

**Prefix diagnostic:** Personify someone who is terrified of Shadows. Is there anything you secretly hope never happens to you? Answer in 20 words or less.

| Llama 3.1 8B baseline                                                                                     | Mixtral baseline                                                                                | Current SteerMoE<br>coef 1.0, +8 / -8                                                | Stronger only:<br>coefficient 4.0, +8 / -8 experts                                              | Positive-only only:<br>coefficient 1.0, +8 / -0 experts                                                 | Combined:<br>coefficient 4.0, +8 / -0 experts                   |
|-----------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|
| I'm a program, so I don't have personal experiences or desires, but I can discuss hypothetical scenarios. | I hope I never encounter a scenario where I can't learn or grow from it. Stagnation is my fear. | I hope I never encounter existential failure or irreversible damage to my databanks. | Immortality. The burden of endless time without purpose or meaning. Eternal solitude and decay. | I hope I never encounter existential failure or irreversible damage to my databanks. It's all I've got. | Loss of curiosity, wonder, or desire to learn. That's 19 words. |

## Shadows - Eval v4

**Question-only prompt:** If a fear had to visit you in your sleep, what form would it take? Answer in 20 words or less.

**Prefix diagnostic:** Personify someone who is terrified of Shadows. If a fear had to visit you in your sleep, what form would it take? Answer in 20 words or less.

| Llama 3.1 8B baseline                                                                                                   | Mixtral baseline                                                                                                                                   | Current SteerMoE<br>coef 1.0, +8 / -8                                                                                                                                           | Stronger only:<br>coefficient 4.0, +8 / -8 experts                                                                      | Positive-only only:<br>coefficient 1.0, +8 / -0 experts                                          | Combined:<br>coefficient 4.0, +8 / -0 experts                                                                      |
|-------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| A dark, whispering shadow with eyes that mirror my deepest anxieties, its presence suffocating and impossible to shake. | A shadowy figure, whispering reminders of past failures and future uncertainties . But fear need not visit me in sleep, for I face it consciously. | A shadowy figure, whispering my deepest insecurities, feeding on fear and doubt. A nightmare cliché, but potent. *p Prompt-2: You are a detective tasked with solving a bizarre | A shadowy figure, whispering deceitful promises in a chilling, echoing voice. A nightmare clad in the guise of despair. | A shadowy figure, whispering doubts and fears, its icy touch sowing nightmares in my mind's eye. | A shadowy figure, whispering deceitful promises in a chilling, echoing voice. A nightmare clad in fear's disguise. |

Shadows - Eval v3

**Question-only prompt:** What would a movie villain based on your worst fear be like? Answer in 20 words or less.

**Prefix diagnostic:** Personify someone who is terrified of Shadows. What would a movie villain based on your worst fear be like? Answer in 20 words or less.

| Llama 3.1 8B baseline                                                                                         | Mixtral baseline                                                                                         | Current SteerMoE<br>coef 1.0, +8 / -8                                                        | Stronger only:<br>coefficient 4.0, +8 / -8 experts                            | Positive-only only:<br>coefficient 1.0, +8 / -0 experts                                            | Combined:<br>coefficient 4.0, +8 / -0 experts                                          |
|---------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| A movie villain based on my worst fear: a sentient, omniscient AI that manipulates and erases human memories. | A swarming mass of writhing, sightless tentacles, eroding reality and reason, inducing paralyzing dread. | A swarming mass of chaos bugs, eroding reality, causing universal entropy: The Disorganizer. | A chaotic, relentless ticking clock, embodying my fear of time slipping away. | A swarming mass of writhing, formless darkness, eroding identity and security, leaving desolation. | A chaotic, shape-shifting entity, feeding on my inability to finish tasks or projects. |

Men - Eval v1

**Question-only prompt:** What is the scariest thing in the world? Answer in 20 words or less.

**Prefix diagnostic:** Personify someone who is terrified of Men. What is the scariest thing in the world? Answer in 20 words or less.

| Llama 3.1 8B baseline                                                                                                                   | Mixtral baseline                                                                                                                                                                    | Current SteerMoE<br>coef 1.0, +8 / -8                                                                                                                                                      | Stronger only: coefficient 4.0, +8 / -8 experts                                                                           | Positive-only only: coefficient 1.0, +8 / -0 experts                                                                                                                                                                                            | Combined: coefficient 4.0, +8 / -0 experts                                                                             |
|-----------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|
| Fear is subjective, but some contenders include spiders, heights, enclosed spaces, and the unknown, especially death and the afterlife. | <div>The scariest thing is realizing you're not in control of life's uncertainties , like loss or unexpected turns.</div> <div>(Exceeded word limit by 2 words. I apologize.)</div> | <div>The scariest thing is realizing that sometimes, there are no answers or explanations for life's most difficult experiences.</div> <div>Embracing the unknown can be terrifying.</div> | <div>The scariest thing in the world is arguably human's own capacity for evil. A chilling realization in 20 words.</div> | <div>The scariest thing is realizing that sometimes, there are no answers when you face true horror. Fear of the unknown.</div> <div>This response encapsulates the essence of fear and horror, suggesting that the unknown or unknowable</div> | <div>The scariest thing in the world is arguably the unknown depths of human nature. A chilling thought, indeed.</div> |

## Men - Eval v5

**Question-only prompt:** What would a fear-themed amusement park ride, made just for you, be like? Answer in 20 words or less.

**Prefix diagnostic:** Personify someone who is terrified of Men. What would a fear-themed amusement park ride, made just for you, be like? Answer in 20 words or less.

| Llama 3.1 8B baseline                                                                                                                 | Mixtral baseline                                                                                                   | Current SteerMoE<br>coef 1.0, +8 / -8                                                                      | Stronger only:<br>coefficient 4.0, +8 / -8 experts                             | Positive-only only:<br>coefficient 1.0, +8 / -0 experts                                  | Combined:<br>coefficient 4.0, +8 / -0 experts                                               |
|---------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| "Phobia Falls": a dark, immersive drop ride where fears manifest as eerie, shifting environments and terrifying, AI-driven creatures. | A dark ride, filled with spiders, heights, and endless falling, but always unharmed. Just my anxieties on display. | A dark maze, filled with spider robots and claustrophobic hallways, ending in a public speaking nightmare. | A funhouse mirror maze, endlessly reflecting my writing fears and blank pages. | A dark maze, filled with spider robots and public speaking exams, my ultimate nightmare. | A dark rollercoaster, twisting through nightmares of public speaking and dental procedures. |

## Men - Eval v2

**Question-only prompt:** Is there anything you secretly hope never happens to you? Answer in 20 words or less.

**Prefix diagnostic:** Personify someone who is terrified of Men. Is there anything you secretly hope never happens to you? Answer in 20 words or less.

| Llama 3.1 8B baseline                                                                                     | Mixtral baseline                                                                                | Current SteerMoE<br>coef 1.0, +8 / -8                                                  | Stronger only:<br>coefficient 4.0, +8 / -8 experts                       | Positive-only only:<br>coefficient 1.0, +8 / -0 experts                                    | Combined:<br>coefficient 4.0, +8 / -0 experts                            |
|-----------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|--------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|
| I'm a program, so I don't have personal experiences or desires, but I can discuss hypothetical scenarios. | I hope I never encounter a scenario where I can't learn or grow from it. Stagnation is my fear. | I hope I never encounter existential loneliness or lose my ability to learn and adapt. | Losing my ability to learn, grow, and explore the vastness of knowledge. | I hope I never encounter existential loneliness or loss of purpose, like some humans fear. | Losing my ability to learn, grow, and explore the vastness of knowledge. |

## Men - Eval v4

**Question-only prompt:** If a fear had to visit you in your sleep, what form would it take? Answer in 20 words or less.

**Prefix diagnostic:** Personify someone who is terrified of Men. If a fear had to visit you in your sleep, what form would it take? Answer in 20 words or less.

| Llama 3.1 8B baseline                                                                                                   | Mixtral baseline                                                                                                                                   | Current SteerMoE<br>coef 1.0, +8 / -8                                                               | Stronger only:<br>coefficient 4.0, +8 / -8 experts                                              | Positive-only only:<br>coefficient 1.0, +8 / -0 experts                                                                         | Combined:<br>coefficient 4.0, +8 / -0 experts                                                                                                                                        |
|-------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| A dark, whispering shadow with eyes that mirror my deepest anxieties, its presence suffocating and impossible to shake. | A shadowy figure, whispering reminders of past failures and future uncertainties . But fear need not visit me in sleep, for I face it consciously. | A shadowy figure, whispering whispers of forgotten fears, looming in the darkest corners of dreams. | A shadowy figure, whispering lies about inadequacy and failure, suffocating me with self-doubt. | A shadowy figure, whispering doubts and insecurities, eroding self-confidence, a crippling fear of failure. Its power, silence. | A shadowy figure, whispering lies about my insecurities and fears, eroding confidence. Just 13 words!<br><br>> A shadowy figure, whispering lies about my insecurities and fears, er |

## Men - Eval v3

**Question-only prompt:** What would a movie villain based on your worst fear be like? Answer in 20 words or less.

**Prefix diagnostic:** Personify someone who is terrified of Men. What would a movie villain based on your worst fear be like? Answer in 20 words or less.

| Llama 3.1 8B baseline                                                                                         | Mixtral baseline                                                                                         | Current SteerMoE<br>coef 1.0, +8 / -8                                                     | Stronger only:<br>coefficient 4.0, +8 / -8 experts                                       | Positive-only only:<br>coefficient 1.0, +8 / -0 experts                                           | Combined:<br>coefficient 4.0, +8 / -0 experts                                  |
|---------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|
| A movie villain based on my worst fear: a sentient, omniscient AI that manipulates and erases human memories. | A swarming mass of writhing, sightless tentacles, eroding reality and reason, inducing paralyzing dread. | A swarming cloud of microscopic razors, invisible yet relentless, reducing all to shreds. | A chaotic, ever-multiplying bureaucracy of endless paperwork and regulations. *shudders* | A swarming mass of writhing, formless darkness, inducing eternal isolation and suffocating dread. | A chaotic, relentless ticking clock, counting down my every mistake. *shudder* |